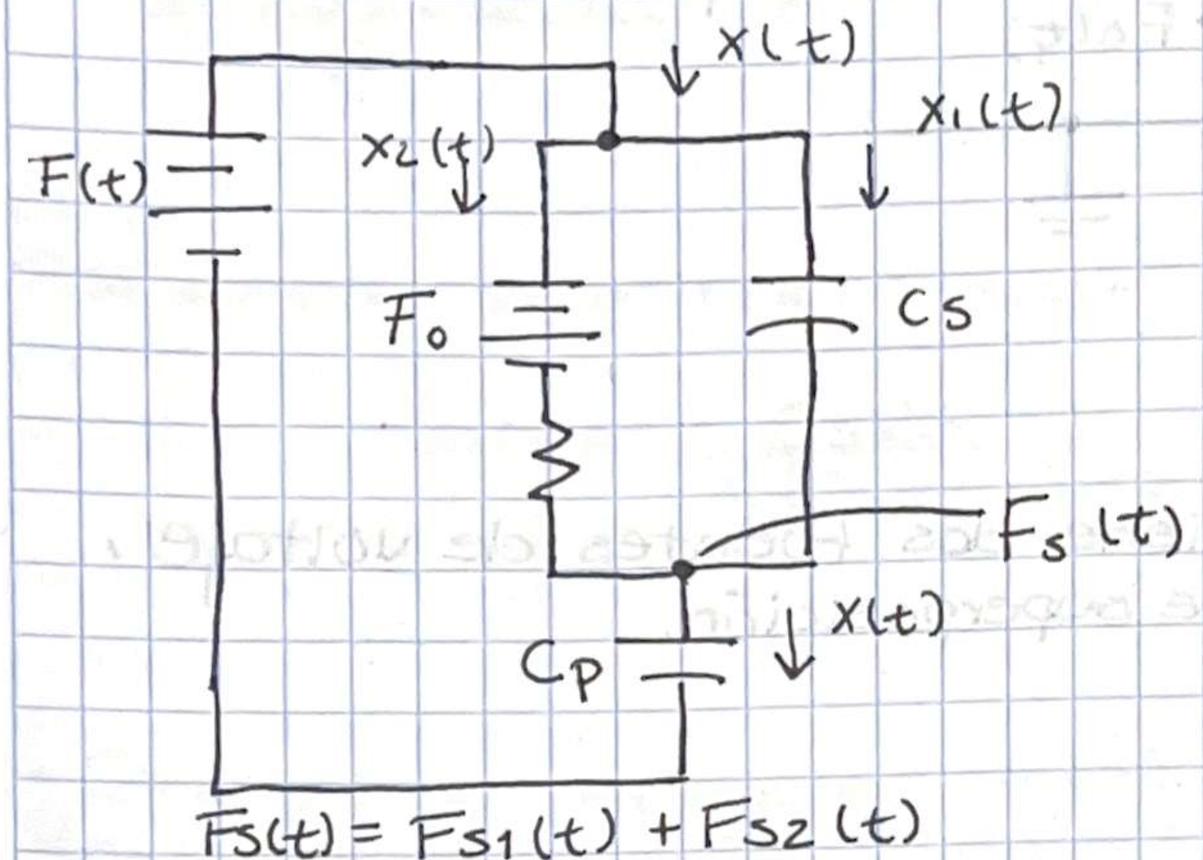
Práctica 3. 26-Oct.-26

Modelado y control de sistema musculoesqueletico.

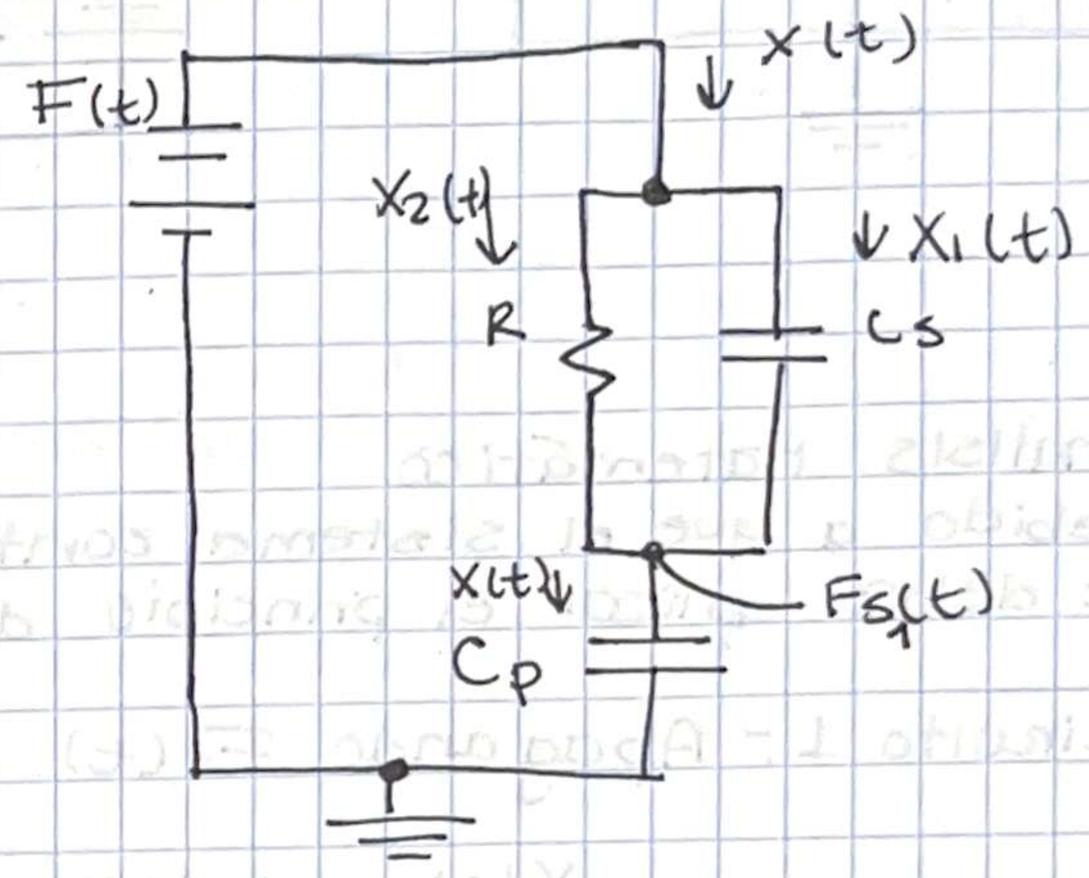
Circuito original

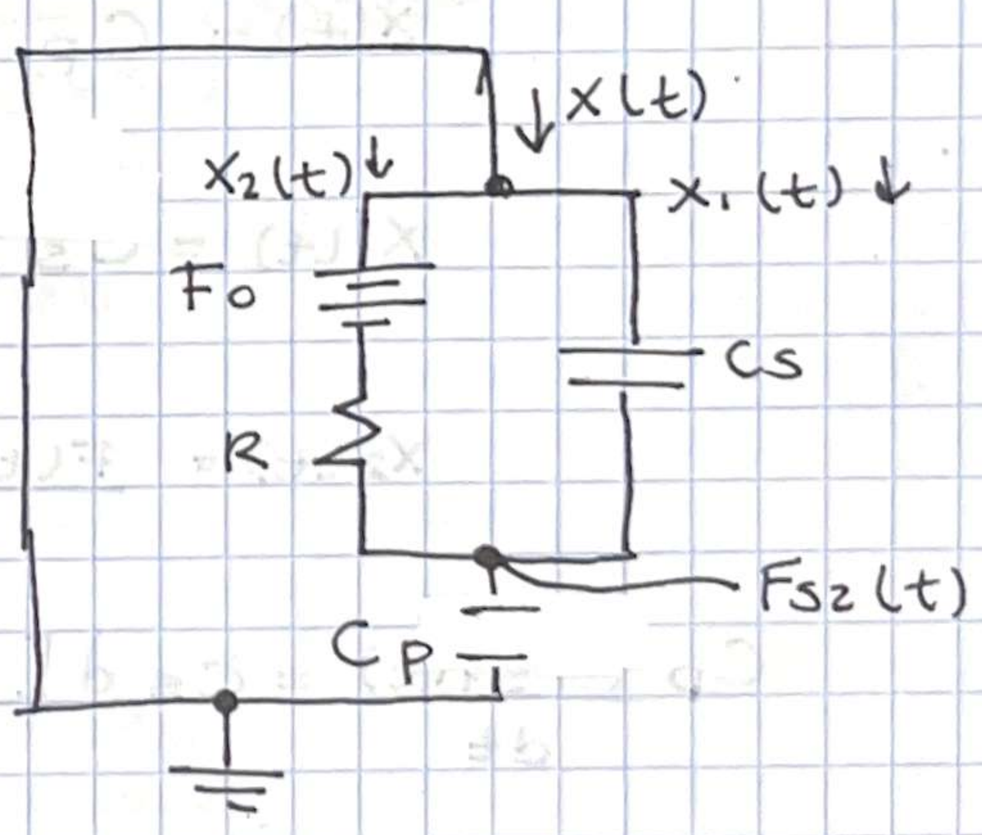
$X(t) = X_1(t) + X_2(t)$

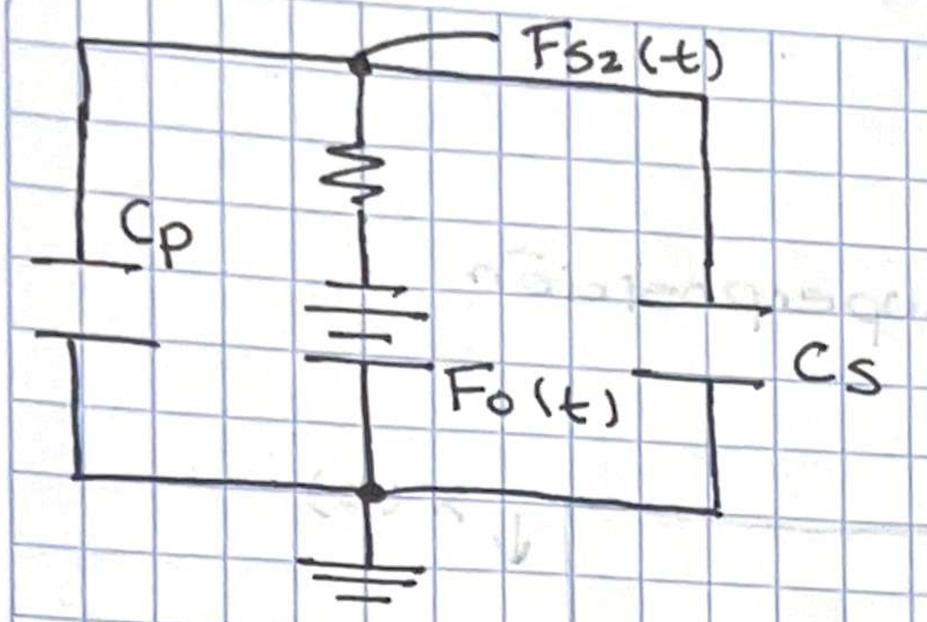


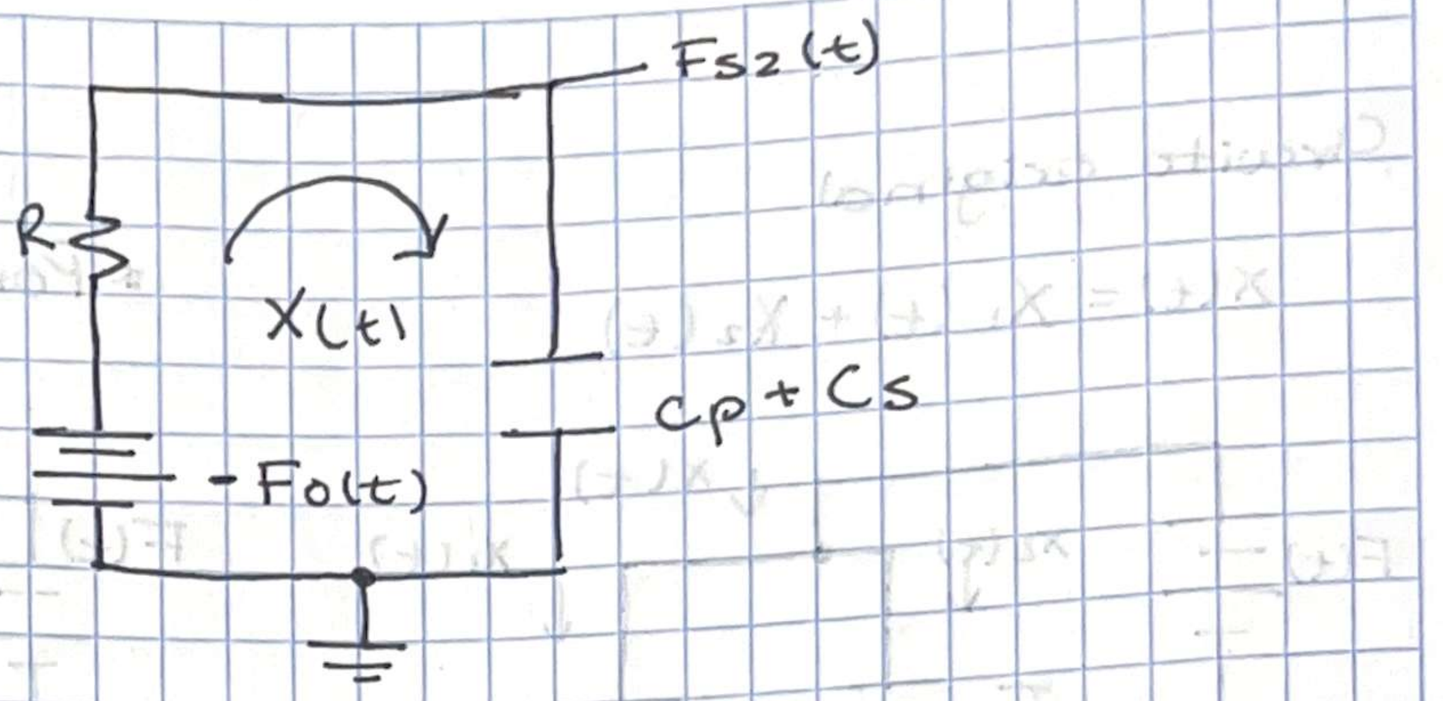
$F_S(t) = F_{S1}(t) + F_{S2}(t)$

* Por superposición









Analisis Matemático

Debido a que el sistema contiene dos fuentes de voltaje, se debe aplicar el principio de superposición.

- Circuito 1: Apagando $F_o(t)$

$$X(t) = X_1(t) + X_2(t)$$

$$X(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$X_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{S1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

Aplicamos L'aplace

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

Agrupamos

$$C_p s F_{s1}(s) + C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

Factorizamos

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s_1}(s) = \frac{C_s R s + 1}{R} F(s)$$

Así queda, la primera salida.

$$F_{s_1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

- Circuito 2.- Apagando $F(t)$.

Recordando que $F_o(t) = \alpha F(t)$

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{s_2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

Aplicamos Laplace.

$$-\alpha F(s) = R X(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$F_{s_2}(s) = \frac{X(s)}{(C_p + C_s)s} \rightarrow \text{salida}$$

$$F_s(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_p + C_s)s}\right) X(s)$$

$$F_s(s) = -\frac{1}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s} X(s)$$

Función de transferencia.

$$\frac{F_{S2}(S)}{F(t)} = -\frac{\alpha}{R(C_p+C_s)s+1}$$

$$F_{S2}(S) = -\frac{\alpha F(t)}{R(C_p+C_s)s+1}$$

- Función de transferencia.

$$F_S(s) = F_{S1}(s) + F_{S2}(s) = \frac{C_s R s+1}{R(C_p+C_s)s+1} F(s) - \frac{\alpha F(s)}{R(C_p+C_s)s+1}$$

$$F_S(s) = \frac{C_s R s+1-\alpha}{R(C_p+C_s)s+1} F(s)$$

$$\frac{F_S(s)}{F(s)} = \frac{C_s R s+1-\alpha}{R(C_p+C_s)s+1}$$

- Error en estado estacionario.

$$e(s) = \lim_{s\to 0} sF(s)\left(1 - \frac{F_S(s)}{F(s)}\right)$$

$$e(s) = s\frac{1}{s}\left(1 - \frac{RC_s s+1-\alpha}{(RC_p+RC_s)s+1}\right)$$

$$e(s) = \alpha$$

- Estabilidad en lazo abierto

$$(RC_p + RC_s)s + 1 = \emptyset$$

$$\lambda = -\frac{1}{RC_p + RC_s}$$

el sistema es estable.